# Three-Point Finite-Difference Differentiation

Plus Composite Trapezoidal and Simpson Integration

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- Approximate first derivatives from sampled data using three-point stencils.
- Validate central, forward, and backward formulas against analytic derivatives.
- Validate composite trapezoidal and Simpson rules on an integral with exact value $\pi$.
- Apply the differentiation workflow to estimate $g$ from free-fall position-time measurements.

- Central:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

- Forward:

$$f'(x) \approx \frac{-3f(x) + 4f(x+h) - f(x+2h)}{2h}$$

- Backward:

$$f'(x) \approx \frac{f(x-2h) - 4f(x-h) + 3f(x)}{2h}$$

- Test functions: $\sin(x)$ at $\pi/4$, $e^x$ at 0.3, and $x^3 - 2x^2 + x - 5$ at 1.2.
- Step size: $h = 10^{-5}$.
- All 9 method-function combinations passed tolerance with errors below $1.5 \times 10^{-10}$.

Differentiation Summary

| method | count | pass_count | max_abs_error | mean_abs_error |
|---|---|---|---|---|
| central | 3 | 3 | 1.259748e-10 | 5.389444e-11 |
| forward | 3 | 3 | 1.404787e-10 | 6.823009e-11 |
| backward | 3 | 3 | 1.404787e-10 | 7.933232e-11 |

Figure 1: Per-method differentiation summary reused from generated Unit 03 artifacts.

- Trapezoidal:

$$\int_a^b f(x)\,dx \approx h\left[\frac{f(x_0)+f(x_n)}{2}+\sum_{i=1}^{n-1} f(x_i)\right]$$

- Simpson (even $n$):

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\left[f(x_0)+f(x_n)+4\sum_{i=1,\,\text{odd}}^{n-1} f(x_i)+2\sum_{i=2,\,\text{even}}^{n-2} f(x_i)\right]$$

- Benchmark integral:

$$\int_0^1 \frac{4}{1+x^2}\,dx = \pi$$

- Subinterval sweep: $n = 4, 8, 16, 32, 64, 128, 256$ for both methods.
- Trapezoidal observed order: 1.99999 (expected 2).
- Simpson observed order: 6.19026 (expected 4; inflated by near-zero terminal errors).
- Finest-mesh errors: trapezoidal $2.543 \times 10^{-6}$, Simpson 0 (within machine precision).

| Method | Expected order | Observed order | Final tolerance pass |
|---|---|---|---|
| Trapezoidal | 2 | 1.99999 | Yes |
| Simpson | 4 | 6.19026 | Yes |

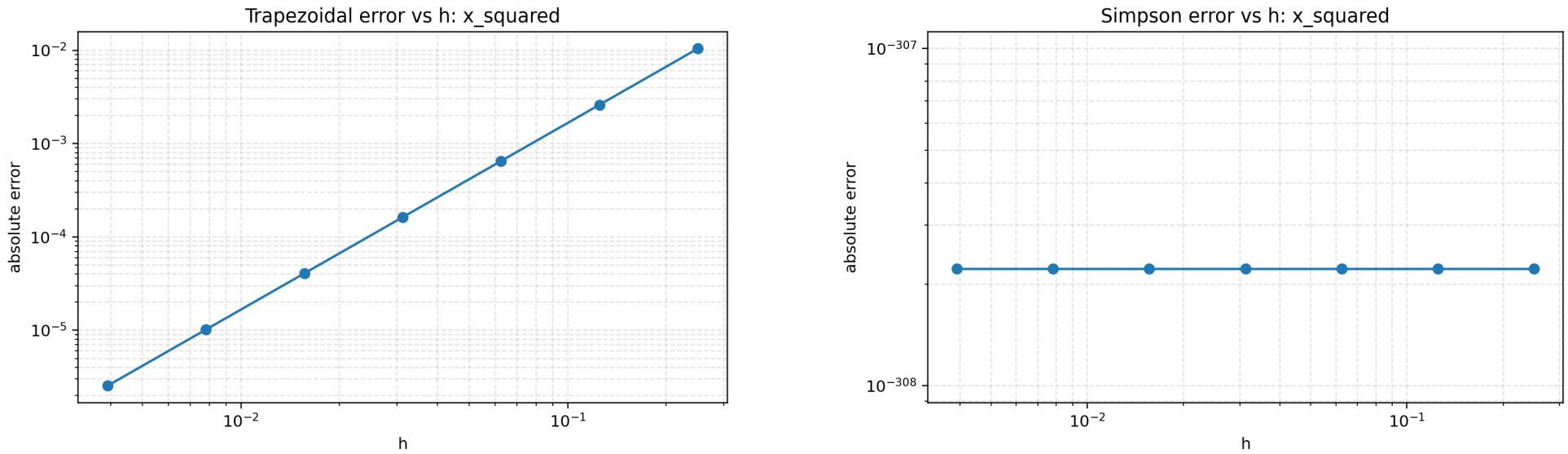


Figure 2: Log-log absolute error trends for composite trapezoidal and Simpson rules on the $\pi$ benchmark.

- Fit a quadratic interpolant to 7 measured position-time samples.
- Differentiate numerically twice using the central stencil.
- Evaluate at $t_0 = 0.174306\,\mathrm{s}$ to recover acceleration.

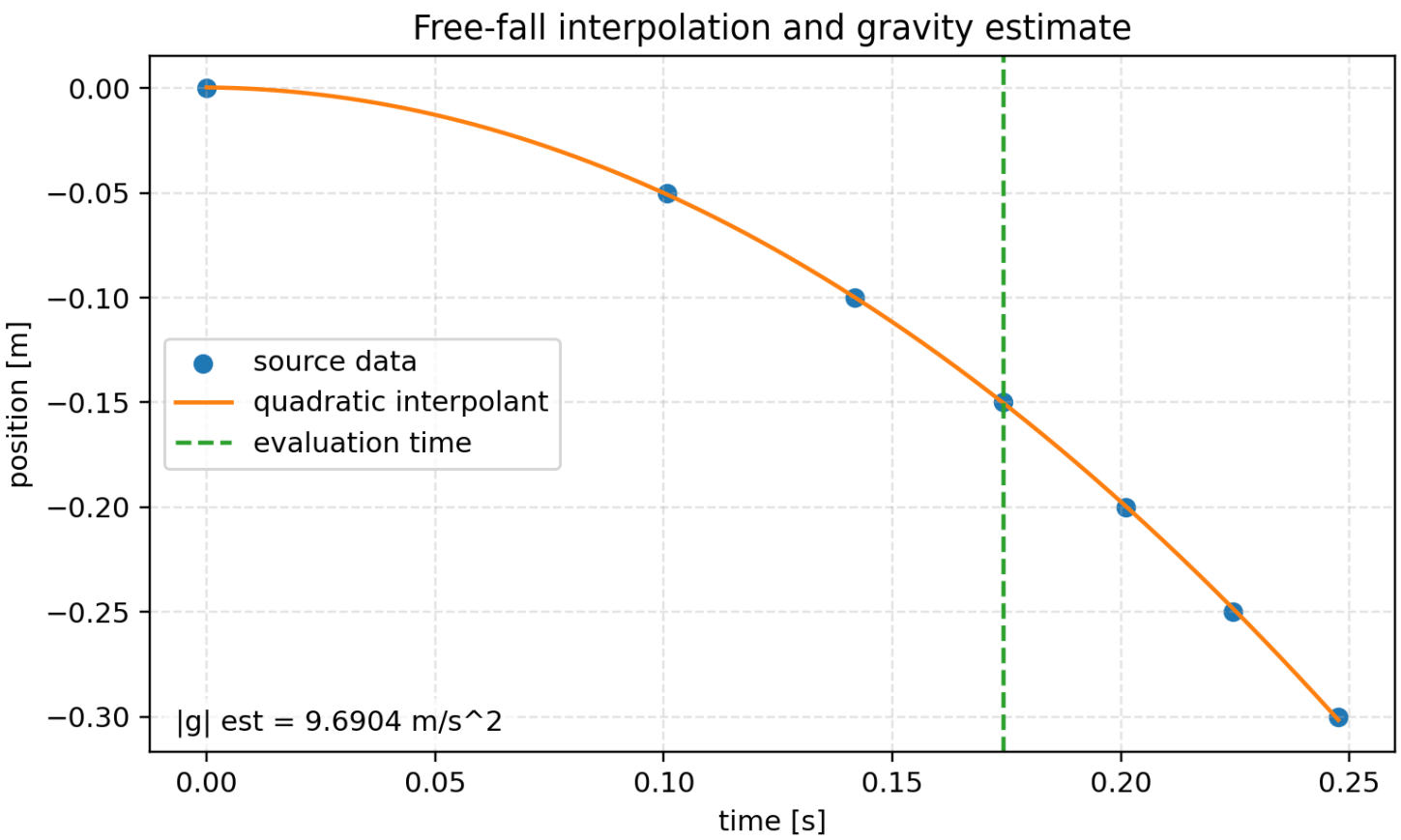

Figure 3: Quadratic interpolant fitted to measured free-fall data.

- Signed estimate: $\hat{a} = -9.6904\,\mathrm{m/s^2}$ (downward direction).
- Magnitude: $|\hat{g}| = 9.6904\,\mathrm{m/s^2}$.
- Reference: $9.81\,\mathrm{m/s^2}$.
- Absolute magnitude error: $0.1196\,\mathrm{m/s^2}$.
- Tolerance check: pass ($0.1196 < 0.15$).

- Three-point finite-difference methods achieved strong analytic accuracy.
- Central differencing provided the best overall error performance.
- Trapezoidal and Simpson integration both passed convergence and final-accuracy checks.
- Simpson reached near machine precision on the smooth benchmark integral.
- Interpolation plus numerical differentiation recovered $g$ within 1.2% of the standard value.

**Questions?**